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ABSTRACT

We examine whether the audit regulation of disclosing key audit matters (KAM) provides value-relevant information to short sellers as informed investors. The theoretical underpinning for examining short sellers' ability and incentives to use KAM disclosures in their stock valuation implications is based on a prediction theory and a skilled information processing theory of short sellers. Using a sample of expanded auditor's reports from UK-listed firms during the 2010–2017 period and hand-collecting a dataset of KAM disclosures, we find no evidence that the short interest is different for the period before than after the U.K.'s expanded auditor's report regulation. However, in our cross-sectional tests, we find that KAM disclosures have a marginal effect on short interest and a positive association between short interest and unexpected and severe KAM disclosures. We conclude that, except for severe KAM that is value-relevant to sophisticated investors, the disclosures in the expanded auditor's report have no valuation implications for short sellers. Our results are robust in examining the reactions of the financial market and analysts to KAM disclosures and addressing potential endogeneity concerns.

1. Introduction

Global auditing standard setters have taken initiatives to enhance the information content and value relevance of the audit report beyond the conventional “pass-or-fail” model by requiring auditors to communicate information on audit scope, financial reporting quality, materiality thresholds, and key/critical audit matters (KAM/CAM) to investors (FRC, 2013; IAASB, 2016; PCAOB, 2017a).¹

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¹ The International Auditing and Assurance Standards Board (IAASB) refers to the expanded auditor's report disclosures as key audit matters (KAM), whereas the PCAOB considers such disclosures as critical audit matters (CAM). In this study, we refer to the expanded auditor's report disclosures as KAM since we use UK audit reports. There are some similarities and differences between KAM and CAM disclosure as both have a similar purpose of communicating audit specific (key or critical) matters to investors beyond the conventional audit reports. However, the process of disclosing KAM is different from that of CAM.

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Prior studies (e.g., Burke, Hoitash, Hoitash, & Xiao, 2023; Burke et al., 2023; Gutierrez, Minutti-Meza, Tatum, & Vulcheva, 2018; Klevak, Livnat, Pei, & Suslava, 2020, 2023; Lennox, Schmidt, & Thompson, 2022) find insignificant and often mixed results of capital market reactions to the expanded auditor's report disclosures. Velte and Issa (2019) synthesize 49 related studies and conclude that results are inconsistent, inconclusive, and mixed. Mashayekhi, Dolatzarei, Faraji, and Rezaee (2023) present mapping the state of expanded audit reporting, including KAM research, and find a gap in the KAM literature. We attempt to void this gap in the KAM literature regarding inconclusive results pertaining to the value-relevance of KAM disclosures by examining whether short sellers, as sophisticated and informed investors, consider the expanded auditor's report disclosures in their investment decisions. This examination is important for future KAM/CAM reporting that may trigger potential market reactions and valuation considerations. Short sellers are regarded as sophisticated investors with a superior ability to access and interpret financial and audit information (Christophe, Ferri, & Hsieh, 2010; Lee, Ling, & Rezaee, 2023). We focus on the 2013 adoption of the expanded auditor's report in the United Kingdom (UK) to test whether expanded auditor's report information is incorporated into informed investors' investment decisions.

The UK adoption is particularly interesting because it is the first regulation where the expanded auditor's report was mandatorily adopted. Thus, it provides a unique setting and sufficient data to examine the market valuation of KAM disclosures. The surprising adoption of the expanded auditor's report for the market in the UK provides us with a setting where we can directly test whether the new disclosures have been incorporated into sophisticated investors' investment decisions. Our study is related to but distinct from prior studies (Gutierrez et al., 2018; Lennox et al., 2022) that examine long and short-window market reactions to the expanded auditor's report. Our study differs from these two related papers in several ways. First, we examine KAM disclosers, whereas these studies focus on disclosing the risks of material misstatements (RMM) included in the expanded auditor's report. Second, we investigate the short sellers' market valuation of KAM disclosures, whereas these studies test market reaction to RMM. Finally, the issue of whether short sellers' responses to unexpected and severe KAM disclosures has not been addressed in prior studies.

It is unclear *priori* whether and how short sellers integrate KAM disclosures into their investment decisions. We argue that short sellers focus largely on material extraordinary and severe information regarding firm prospects (irregularities, abnormalities, future performance), as suggested by Lee et al. (2023). Thus, short sellers are likely to find the KAM information relevant in highlighting audit risk areas and KAM severe. It is possible that short sellers can acquire information about a firm's future financial performance from other sources through their investigation and predictive ability processes (Jain, Jain, & Rezaee, 2016). It is also possible that short sellers can foresee the information in the expanded auditor's report and front-run KAM disclosures (Lee et al., 2023). Therefore, the expanded auditor's report and related KAM disclosures may not have any valuation implications for short sellers. These possibilities introduce tension in our research questions: "Do the expanded auditor's report disclosures provide value-relevant information to short sellers?" and "Do unexpected and severe KAM disclosures have investment implications for short sellers?" Therefore, it is unclear whether we would observe any reactions by short sellers to the expanded auditor's report and/or KAM disclosures.

Using a sample of expanded auditor's reports from UK-listed firms from 2010 to 2017 and hand-collecting a dataset of KAM disclosures, we find that short interest is *not* significantly associated with the 2013 adoption of the expanded auditor's report. Our cross-sectional tests in the post-2013 period further indicate that the attributes of the expanded auditor's report have a largely insignificant effect on the short interest. However, we find marginal evidence that short sellers incorporate the information provided by the number of KAM, particularly the unexpected and severe KAM disclosures, into their investment decisions.² We also examine the association between the severity of KAM and short interest and find that short interest is positively associated with both unexpected and severe KAM. We conclude that, except for KAM information, other disclosures in the expanded auditor's report have largely no valuation implications for short sellers. Our results are robust to additional analyses to examine the reactions of the financial market and analysts to KAM disclosures and to address potential endogeneity concerns.

Overall, we find (1) no evidence that the short interest is different for the period before and after the U.K.'s expanded auditor's report regulation, which is consistent with the findings of Gutierrez et al. (2018), Lennox et al. (2022), and Burke et al. (2023); (2) some evidence that KAM disclosures have a marginal effect on short interest; and (3) short interest is positively associated with both unexpected, severe and impactful KAM disclosures. To the best of our knowledge, findings 2 and 3 of the above are not being addressed in the literature, and thus, our study makes several contributions. First, our findings suggest that short sellers already consider the information provided by auditors before publicly disclosing the audit report and only react to unexpected and severe KAM disclosures. These results are consistent with those of Burke et al. (2023:61) that find "a negative market reaction when the number of CAMs disclosed deviates from expectations and when CAMs are disclosed in subject areas where they are not expected." Thus, investors and audit clients should be interested in our findings, given the value relevance of unexpected and severe KAM disclosures that suggest these disclosures are riskier and severe than anticipated. Audit firms may want to consider our evidence in disclosing material, unexpected, and severe KAM/CAM disclosures in their future audit reports. Second, our results contribute to the global debate on the value relevance of KAM disclosures (e.g., Bedard, Gonthier-Besacier, & Schatt, 2019) by showing that only unexpected and severe KAM disclosures exhibit value relevance in providing incremental information to capital markets. Our results contribute to this line of research by providing marginal support for the extent of KAM and stronger support for unexpected and severe KAM disclosures that investors and auditors view as relevant and material.

Third, our findings support standard setters' efforts to enrich the informational content of the current audit report (IAASB, 2016; Public Company Accounting Oversight Board, 2017b). Our results suggest that those in charge of a company's corporate governance,

² We define unexpected KAM as the residual of a regression of KAM on their normal determinants and severe KAM are the ones that relate to (i) fraud, (ii) going concern, or (iii) management override of internal controls.

particularly the audit committee, should oversee KAM disclosures, as [Rousseau \(2021\)](#) finds that the audit committee's cognitive diversity in knowledge, abilities, and expertise is positively related to KAM quality. Finally, our results suggest that, in the preparation of future audit reports, auditors should focus on material and severe KAM/CAM that address the quality of financial reports, the challenges involved in the preparation of high-quality financial statements, and the areas that are more complex and severe to audit such as business continuity, going concern, risk assessment, and internal controls. These results are consistent with those of [Burke et al. \(2023\)](#) and [Klevak et al. \(2020\)](#), which suggest variations in the number and textual disclosed CAMs are associated with abnormal returns and audit fees. Our results suggest that severe KAM/CAM can reveal audit-specific risk information and thus generate market reactions.

The rest of the paper is structured as follows: Section 2 describes the UK institutional background and literature review. Section 3 provides a theoretical framework and hypothesis development. Section 4 provides our research methodology, while Section 5 presents the results of empirical tests. The last section concludes and provides suggestions for future research.

2. Institutional background and literature review

2.1. Institutional background

In June 2013, the Financial Reporting Council (FRC) in the UK released International Standard on Auditing (ISA) 700, "The Independent Auditor's Report on Financial Statements," which requires that auditors present the expanded auditor's report for fiscal years ending on or after September 30, 2013, and onwards ([FRC, 2013](#)). According to the ISAs ([ISA 700, n.d.](#), paragraphs 8 and 39), KAM disclosures are "[T]hose matters that, in the auditor's professional judgment, were of most significance in the audit of the financial statements of the current period." The IAASB, in September 2014, issued the International Standard on Auditing ([ISA 701, n.d.](#)) No. 701, which requires auditors to communicate KAM disclosures for audits of listed companies ([IAASB, 2016](#)). Although the number of KAM communicated in the audit report may depend on the firm's complexity, the nature of its business and environment, and the facts and circumstances of the audit engagement, the IAASB expects that the report presents at least one KAM.

In the United States, on June 1, 2017, the PCAOB issued the auditing standard ([Public Company Accounting Oversight Board, 2018](#), AS 3101), which should be fully implemented by large, accelerated filers for fiscal years ending on or after June 30, 2019 ([Public Company Accounting Oversight Board, 2019](#)). For other companies, the effective date is December 15, 2020 ([PCAOB 2017a](#), AS 3101). According to AS 3101, the expanded auditor's report should include information regarding the auditor's responsibilities, the procedures applied during the audit, and the disclosure of CAMs. While the new auditing standard (AS 3101) retains the conventional pass/fail audit model, it requires auditors to include a discussion of CAM in their reports.³ KAM and CAM disclosures are intended to inform investors about the important audit issues relevant to material risks of financial misstatements, high levels of management judgment, and estimation uncertainty.

2.2. Literature review

2.2.1. Critical/key audit matters

Several studies examining information content and economic consequences of the expanded auditor's report disclosures (e.g., [Burke et al., 2023](#); [Gutierrez et al., 2018](#); [Klevak et al., 2020, 2023](#); [Lennox et al., 2022](#)) find insignificant and often mixed results of capital market reactions to these disclosures. [Velte and Issa \(2019\)](#) provide a synthesis of 49 related studies, and [Mashayekhi et al. \(2023\)](#) present mapping the state of expanded auditor's disclosures. Specifically, [Gutierrez et al. \(2018\)](#), using short-window market tests, and [Lennox et al. \(2022\)](#), employing longer-window tests, document that the new audit disclosures do not bring additional information to the investors, as their tests do not support the view that the adoption of the expanded audit report is associated with capital market reactions. [Porumb, Karaibrahimoglu, Lobo, Hooghiemstra, and De Waard \(2019\)](#) find that, from a debt market perspective, the RMM included in the expanded auditor's report does provide information to banks and impacts adopting firms' loan contracting. [Porumb, Karaibrahimoglu, Lobo, Hooghiemstra, and De Waard \(2021\)](#) find an association between the expanded auditor's report and reduced loan spread and longer maturity for loan facilities of adopted UK firms compared to non-adopted firms. [Kelton and Montague. \(2018\)](#) address the unintended consequences of uncertainty disclosures (e.g., CAM issues) made by auditors and managers on nonprofessional investor judgments.

A stream of research focuses on the value-relevance of KAM/CAM disclosures. For example, [Sirois, Bedard, and Bera \(2018\)](#) document that CAM disclosures have attention-directing impacts, but investors perceive audit reports as less informative when an auditor discloses multiple CAMs. [Carcello \(2012\)](#) argues that investors benefit from the disclosure of CAMs and thus support auditors' communication of such information. Other studies provide mixed evidence of the impact of CAM disclosures on auditors' liability. [Christensen et al., 2014](#), by experimenting with nonprofessional investors (business school graduates), find that investors who receive CAM information are more likely to modify their investment decisions than those who do not. [Ernst & Young \(2013\)](#) and [Gaetano \(2014\)](#) state that KAM disclosures could increase litigation against auditors, whereas [Brasel, Doxey, Grenier, and Reffett \(2016\)](#) find that KAM disclosures could decrease auditors' liability subject to the type of material misstatements. Other studies find that KAM disclosures are associated with an improvement in financial reporting quality ([Reid, Carcello, Li, & Neal, 2019](#)), and improve stock

³ We contrast KAM and CAM requirements by reviewing recent auditor reports and find no significant differences in standards and practices of both KAM and CAM with one exception that KAM disclosures are often greater in number and length compared to CAM disclosures.

price informativeness by reducing synchronicity (Zhai, Lu, Shan, Liu, & Zhao, 2021), are credible channels for auditors to monitor the disclosure of financial reporting risks associated with goodwill impairment (Andreicovici, Jeny, & Lui, 2023), and KAM disclosures pertaining to mergers and acquisitions (M&A) are predictive of poor post-acquisition performance and caused CEO turnover (Nylen, Wangerin, & Zehms, 2023). Drake, Goldman, Lusch, and Schmidt (2023) find no evidence of the relationship between tax-related CAM disclosures and financial reporting quality or tax avoidance and some evidence of an association between tax-CAM disclosures and tax accrual quality and tax-related earnings management.

Brown, Majors, and Peecher (2014) conclude that CAM disclosures lower perceived auditor liability in litigation resolution settings. Kachelmeier, Schmidt, and Valentine (2014) find that CAM disclosures are a partial disclaimer of auditor responsibility. Dee, Luo, and Zhang (2021) report that firms with CAM reporting exhibit a significant decrease in material weaknesses in their internal control over financial reporting. Bedard et al. (2019) examine the financial market reaction to the justification of assessments (JOA), which is like CAMs, and find disclosures of JOA do not generate any capital market reactions. The quality, cost, and efficiency of the audit are unaffected by such disclosures. Burke et al. (2023) find no market reaction or changes in audit fees and quality associated with CAM disclosures. Klevak et al. (2020) report that firms that disclose more CAMs experience negative market reactions, reduced analyst earnings forecasts, and more stock price volatility. Velte and Issa (2019) conclude that results regarding shareholder reactions to KAM disclosures are inconsistent, inconclusive, and mixed. Camacho-Miñano, Muñoz-Izquierdo, Pincus, and Wellmeyer (2023) argue the predictability of KAM disclosures and find the greater the number of KAM disclosures, the higher the financial distress level of a firm. Overall, the literature on the expanded auditor's report disclosures provides largely inconclusive evidence, and more research is required to assess the information content of CAMs/KAM. Recently, Klevak, Livnat, Pei, and Soslava (2023) found increased perceived uncertainty regarding more volatile stock prices and dispersed analyst forecasts for firms with more extensive CAM disclosures. Lennox et al. (2022) argue that the lack of market reactions to KAM disclosures is not because auditors disclose irrelevant information. Still, rather, investors already know about such disclosures of accounts with high levels of judgement and financial reporting risks prior to KAM reporting. This study contributes to the literature on the value-relevance and informativeness of KAM reporting by examining the association between KAM disclosures and short interest.

2.2.2. Short sellers

Two research streams examine the role of short sellers in the marketplace and how they process information in their investment decisions. The first stream of research suggests that short sellers are informed market participants who process public information in a timelier and more effective manner than average investors because of their skilled information processing ability and trade on such information before it is fully reflected in stock prices (Drake, Rees, & Swanson, 2011; Engelberg, Reed, & Ringgenberg, 2012). For example, Dechow, Hutton, Meulbroek, and Sloan (2001) and Boehmer and Wu (2013) argue that short sellers normally target overvalued stocks that are not in line with their fundamentals. Jain et al. (2016) find a negative link between corporate social responsibility (CSR) measures and short-selling positions, suggesting that short sellers target firms with low CSR ratings.

The second stream of research argues that short sellers can predict a firm's financial challenges and earnings capacity before they are publicly disclosed because of their prediction ability (Christophe et al., 2010). Short sellers target firms with unique financial reporting environments and challenges facilitated by leaked information about poor accounting quality. For example, Diamond and Verrecchia (1987) argue in their theoretical paper that short sellers take advantage of information by selling shares before negative information is available to the public. Christophe et al. (2010) show that short sellers take short positions before analysts release their downgrade recommendations. In addition, several studies document that short sellers predict stock prices and returns (e.g., Asquith, Pathak, & Ritter, 2005; Boehmer, Jones, & Zhang, 2008; Cohen, Diether, & Malloy, 2007; Desai, Ramesh, Thiagarajan, & Balachandran, 2002; Diether, Lee, & Werner, 2009). Khan and Lu (2013) examine whether short sellers trade ahead of or pursuant to inside sales and find evidence of short sellers front-running insider sales suggesting information leakage from insiders to short sellers in firms with poor accounting quality. Akbas, Meschke, and Wintoki (2016) find that sophisticated investors, such as short sellers, better predict future earnings surprises and other firm-specific news outcomes. Lee et al. (2023) find evidence of short sellers' ability to front-run the SEC comment letter disclosures. Our study complements prior research on the role of short sellers in the marketplace related to the expanded auditor's report and KAM disclosures.

3. Theoretical framework and hypotheses development

The theoretical intuition for short sellers' ability and incentives to use KAM disclosures in their stock valuation implications is based on a prediction theory, and a skilled information processing theory of short sellers suggested in prior studies (e.g., Christophe et al., 2010; Engelberg et al., 2012; Jain et al., 2016; Lee & Rezaee, 2022). The prediction ability of short sellers depends on their experiences and skills in analyzing and assessing the specific stocks, market, and relevant information accurately as well as their research, analysis, and access to information. Skilled information processing theory implies that short sellers process and interpret information effectively and efficiently by analyzing information, identifying trends and patterns, and making informed predictions. We posit short sellers are interested in employing KAM disclosures in their investment valuation. On the one hand, short sellers read audit reports carefully to identify material and measurable KAM disclosures and use relevant KAM information to assess risks and opportunities associated with KAM disclosure. On the other hand, short sellers get their investment valuation information from various sources, and KAM disclosures may not be considered important to short sellers.

Short sellers' interest in the audit report is driven by the incentives to target firms with poor financial quality (Bases, Vlastelica, Baldwin, & Bendeich, 2011). The expanded auditor's report is likely to convey incrementally useful information to short sellers relative to the conventional "pass or fail" audit report in several ways. First, the expanded auditor's report is likely to affect short

sellers' predictions of firms' future financial challenges. The extent to which the additional audit-specific disclosures were previously unknown information to short sellers, they will incorporate such information in their investment decisions. Second, the expanded auditor's report may "indirectly update beliefs about the firm's financial reporting quality" (Gutierrez et al., 2018, p. 9), which can have valuation implications for short sellers. Finally, the expanded auditor's report disclosures create an opportunity for short sellers by helping them identify those firms with significant uncertainty and risk (e.g., fair value, revenue recognition, going concern).

Therefore, the newly disclosed information in KAM reporting indirectly influences short sellers' investment decisions. Since the expanded auditor's report information consists of audit scope discussion, materiality thresholds, and KAM, the way the overall report or individual disclosures will be associated with short selling interest is *a priori* unclear. It is also possible that KAM disclosures, by providing more information for all users of audited financial statements, create the same level-playing field that minimizes the information advantage that short sellers have over average investors. It is also possible that short sellers perceive a higher risk of misstatement from KAM disclosures, leading to a positive association between KAM disclosures and short interest. A contrasting perspective is that investors other than short sellers may perceive an improvement in audit quality, internal control, and hence financial reporting reliability in the KAM regime, making KAM disclosures less value-relevant to short-sellers suggesting a negative effect on short interest. Lee et al. (2023) find evidence of short sellers' ability to front-run information included in the SEC comment letters. Lennox et al. (2022) argue that investors were already informed about the RMM information in the financial reporting prior to the disclosure of such information by auditors.

Taken together, it is *ex-ante* unclear whether the expanded auditor's report conveys any valuable and material information to short sellers, which affects their investment decision. These possibilities introduce tension in our research question of whether the expanded auditor's report disclosure is associated with short interest. Thus, to determine the possible lead-lag relation between short interest and the expanded auditor's report disclosures, we formulate our expectations for the overall impact of the 2013 adoption of the expanded auditor's report for short sellers and state our hypothesis (in null form) as follows.

Hypothesis 1. There is no association between short interest and the 2013 adoption of the expanded auditor's report.

The individual impact of expanded auditor's report disclosures in the post-adoption period relevant to KAM disclosures, number of disclosures, and materiality thresholds may be value-relevant for short sellers. Given that inherently challenging information is of the most interest to short sellers, as reported in prior studies (e.g., Boehmer et al., 2008; Desai, Krishnamurthy, & Venkataraman, 2006; Karpoff & Lou, 2010), the disclosure of complex and severe KAM should most likely impact their investment behavior. Aside from KAM, the disclosure of materiality thresholds in the new report is important since it directly indicates auditing quality (Amiram, Chircop, Landsman, & Peasnell, 2017).⁴ Nonetheless, it is also possible that short sellers' front-run auditors disclose KAM and materiality thresholds because of their information processing and predictive abilities (Khan & Lu, 2013; Lee & Ling, 2019).

We argue that short sellers consider KAM disclosures that identify financial areas requiring significant estimation (goodwill impairment, fair value estimation) or are more complex and severe to audit (revenue recognition, long-term contracts, going concern). The severity of disclosed KAM is defined as the number of words discussing KAM, the average number of words discussing KAM, disclosed materiality threshold, disclosed unexpected KAM, and disclosed severe KAM. We posit that more severe KAM present the signals of material financial reporting challenges and complex audits, and thus they should cause more changes in short interest than less severe KAM. We state our next hypotheses as follows.

Hypothesis 2. There is a positive association between short interest and disclosed KAM.

Hypothesis 3. There is a positive association between short interest and the severity of disclosed KAM.

4. Research method

4.1. Sample selection, and variable construction

We hand-collect data regarding the expanded auditor's report-specific measures and attributes including KAM disclosures from the annual reports of premium listed firms on the LSE in 2013–2017. Following Porumb et al. (2021) and Camacho-Miñano et al. (2023), we hand-collect the information regarding the attributes and content of the expanded auditor's report, including the number of KAM disclosures, the number of words in the KAM section of the expanded auditor's report, average words per KAM, the materiality threshold, and the severity of KAM disclosures. We hired a team of five graduate research assistants in accounting at our current institutions. They are asked and guided to thoroughly review each of the expanded auditor's reports from firms' official websites or through Google searches. To ensure accuracy, reliability, and consistency, all research assistants have received a minimum of 4 h of training on identifying, measuring, and assessing the quality of KAMs used in the reports. Each research assistant independently performed the data collection process, and one research assistant was assigned to cross-check the data collection.

Furthermore, we use the Audit Analytics database to collect KAM data for further analyses of analysts and market reactions to KAM disclosures. Additionally, we collect information regarding the audit firm, audit partner identity, experience, educational background, gender, and audit fees from the annual reports. Following Lee et al. (2023), we measure short interest as total shares sold short-scaled by the total shares outstanding at time *t*.

⁴ According to Amiram et al. (2017), a low materiality threshold is associated with an increased reporting quality, as perceived by investors.

We collect short interest data from the Compustat Supplemental Short Interest Database and data on the number of outstanding shares from CRSP. Consistent with prior research (Blau, Fuller, & Van Ness, 2011; Jain et al., 2016; Lee & Ling, 2019), we use two measures of short intensity commonly used in the literature. First, the short intensity ratio (*Short_int*) is calculated as the number of shares shorted (positions held on the 15th business day of each month) divided by the number of outstanding shares in the preceding month. Second, we compute a short intensity ratio adjusted (*Short_int_adj*) measure, calculated as the adjusted number of shares shorted (positions held on the 15th business day of each month) divided by the number of outstanding shares in the preceding month.

We merge short interest data with the information in the expanded auditor's report by matching gvkeys from Compustat with the available ISIN codes of the UK firms. We further collect firm-specific financial data from Worldscope DataStream. Our initial sample starts with UK firms that adopted ISA 700 regarding the expanded auditor's report disclosures in 2013. After eliminating missing variables, we obtained a sample of 2386 firm-year observations from 2010 to 2017 to test Hypothesis 1.⁵ To test Hypothesis 2, we focus solely on the post-2013 period. This filter results in a restricted sample of 1349 observations from 246 firms. Our analyses focused on companies that adopted the 2013 standards, following the framework outlined in Gutierrez et al. (2018). The extended reports centered around the regulatory cut-off date of July 2017. Our analysis considers control observations from companies aligned with the 2013 standards, as detailed in Gutierrez et al. (2018). This subset of observations covers four years, encapsulating the extended reports surrounding the regulatory cutoff in 2017.

We follow prior literature and include control variables potentially associated with short interest and the expanded auditor's report for a firm. Specifically, we control for firm size (*Firm Size*), market-to-book ratio (*MTB*), and trading volume (*Mkt_Cap*) since firms with a larger size, more overpriced stocks, and higher trading volume tend to have higher short interest (Forde, Jones, & Putniņš, 2016; Jain et al., 2016; Kot, 2007). We also control for tangibility (*Tang*), total inventories (*Invent*), loss-making firms (*Loss*), working capital (*WKap*), long-term debt (*LT_debt*), common equity (*BV_Eq*), book value per share (*BPS*), and Big4 auditor (*Big4*). Detailed variable definitions and their data sources are presented in the Appendix.

4.2. Models

Given that short sellers target firms with financial challenges and disclosed KAM addresses financial challenges, we test the possible association between short interest and KAM disclosures after controlling for other determinants of short interest and KAM. To test the predictions of Hypothesis 1, we build on previous literature (e.g., Lee et al., 2023) to construct the following model:

$$\text{Short_int_measure} = \delta_0 + \delta_1 \text{Post} + \delta_i \text{Firm-specific controls(lagged)} + (\text{Industry, Year, Audit firm dummies}) + \varepsilon_t \quad (1)$$

Where *Short_int_measure* represents, sequentially, *Short_int* and *Short_int_adj*, *post* is a dummy variable that takes the value of 1 for financial years after 2013 or 0 otherwise. *Control variables* include *ROA*, *Size*, *LT_Big4*, *Gender BPS Inventory*, *Tangibility*, *BV_Eq*, and *Mkt_Cap*, as defined in Appendix B. After controlling for the other variables, we expect the regression coefficient on *Post* to be positive and significant.

Further, to test our hypothesis of the possible association between the attributes of the expanded auditor's report and short interest, we develop the following model:

$$\text{Short_int_measure} = \delta_0 + \delta_i \text{Expanded-audit report attributes} + \delta_k \text{Firm-specific controls(lagged)} + (\text{Industry, Year, Audit firm dummies}) + \varepsilon_t \quad (2)$$

Where *Expanded-audit report attributes* are, in sequence: (1) *KAM*, the total number of material misstatements disclosed in the expanded auditor's report; (2) *Materiality*, the natural logarithm of the materiality amount disclosed in the expanded auditor's report; (3) *Words_KAM*, the number of words in the expanded auditor's report directly connected to KAM; (4) *Average_Words_KAM*, the number of words in the expanded auditor's report connected to KAM divided by the number of KAM; (5) *Unexpected_KAM* the residual of regression of KAM on their normal determinants; and (6) *Severe_KAM*, an indicator variable equal to 1 for KAM that relate to fraud, going concerned, or management override of internal controls or 0 otherwise. In addition to previously defined firm-specific characteristics, we control for attributes of the expanded auditor's report (*KAM*, *Materiality*, *Words_KAM*, *Average_Words_KAM*, and *Severe_KAM*) when not used as dependent variables. We include *industry, year, and audit firm fixed effects* in our models to control for the potential effect of unobserved heterogeneity. We cluster standard errors by the firm in all estimations.

5. Results

5.1. Descriptive results

In Table 1, Panel A, we present descriptive statistics for the full sample. Moreover, we present in Panel B and Panel C the characteristics of the post- and pre-2013 adoption samples, respectively. The median, minimum, maximum, and standard deviation of all variables used in our analyses for the 2386 firm-years in our sample are shown in Panel A. We further show, in Table 2, the Pearson correlation matrix. The correlations between the variables considered in our tests are generally under 0.7, suggesting that

⁵ We follow DeFond et al. (2015) and Gutierrez et al. (2016), and in the test of Hypothesis 1, we use a -2-to-+2-year window around the adoption. We choose to do this to decrease the effect of other potential confounding events.

multicollinearity between independent variables is unlikely to be an issue in our data.

5.2. Empirical results

5.2.1. The expanded Auditor's report

We present the results of testing [Hypothesis 1](#) in [Table 3](#). The coefficient of our main variable of interest, *Post*, is insignificant in Models 1 and 2. This is consistent with the overall adoption of the expanded auditor's report not impacting the investing behavior of short sellers. This test assesses the joint impact of all disclosures in the expanded auditor's report. The sign and significance of control variables are generally consistent with prior literature.

5.2.2. Key audit matters

We estimate Equation (3) for our post-adoption sample, focusing on the coefficient of the KAM variable. Results presented in [Table 4](#) show that the coefficient of KAM is positive and marginally significant at the 10 percent level. These results suggest that short sellers regard KAM to provide value-relevant information and are likely to consider such disclosures informative and incorporate them into their investment decisions. Our results in [Tables 5 and 6](#) indicate that the coefficients of *Words_KAM* and *Average_Words_KAM* are insignificant in all estimations. This is in line with short sellers not integrating the explanations of auditors in their valuations ([Lee & Rezaee, 2022](#)). We find that the coefficients of *Maturity* in [Table 7](#) are insignificant in all estimations. This result aligns with the

Table 1

Descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
Panel A – Full data					
<i>Short_int</i>	2386	0.1649438	0.6986141	−0.2665218	6.267337
<i>Short_int_adj</i>	2386	0.1710022	0.7120961	−0.2791279	6.364257
<i>KAM</i>	2386	5.289606	1.531621	2	9
<i>ROA</i>	1669	0.0421445	0.9935739	−2.233932	6.589102
<i>AFee</i>	1495	−0.0152137	1.034434	−0.6659454	4.06378
<i>Size</i>	2336	0.0177819	1.03686	−0.3002843	6.521611
<i>Invent</i>	1694	−0.0087671	1.044129	−0.4771927	6.357436
<i>W_kap</i>	1694	−0.0282667	1.049279	−4.285716	2.578948
<i>LT_debt</i>	1694	0.0202985	1.050917	−0.3369235	4.794565
<i>Common</i>	1694	0.0253935	1.05027	−0.2912417	4.531703
<i>BPS</i>	1694	0.0296092	1.018001	−0.86465	4.213905
<i>Big4</i>	2386	0.9484493	0.2211645	0	1
<i>Gender</i>	2141	0.9425502	0.2327544	0	1
<i>Mkt_Cap</i>	1570	0.0234027	1.054418	−0.3441499	5.016553
Panel B – Post data					
Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Short_int</i>	1938	0.0710553	1.091761	−0.2699581	49.37123
<i>Short_int_adj</i>	1938	0.0506515	1.084473	−0.2815721	50.18378
<i>KAM</i>	2386	5.289606	1.531621	2	9
<i>ROA</i>	1481	0.0384037	1.010268	−2.650318	6.589102
<i>AFee</i>	1357	0.0583721	1.074599	−0.6659454	4.06378
<i>Size</i>	2000	0.0297608	1.092956	−0.3002843	6.521611
<i>Invent</i>	1506	0.004576	1.057799	−0.4771927	6.357436
<i>W_kap</i>	1506	−0.0565947	1.108181	−4.285716	2.578948
<i>LT_debt</i>	1506	0.064258	1.10705	−0.3369235	4.794565
<i>Common</i>	1506	0.0596245	1.109109	−0.2912417	4.531703
<i>BPS</i>	1506	0.0019862	1.024676	−0.86465	4.213905
<i>Big4</i>	2025	0.9516049	0.2146526	0	1
<i>Gender</i>	1755	0.9441595	0.2296788	0	1
<i>Mkt_Cap</i>	1407	0.0575182	1.108273	−0.3441499	5.016553
Panel C – Pre data					
Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Short_int</i>	448	−0.0326137	0.9531629	−0.2699581	145.3244
<i>Short_int_adj</i>	448	−0.0232487	0.9578426	−0.2815721	66.37209
<i>KAM</i>	0	0	0	0	0
<i>ROA</i>	374	−0.1520746	0.9443188	−2.233932	1.223802
<i>AFee</i>	324	−0.2444781	0.528312	−0.635431	1.156429
<i>Size</i>	522	−0.114026	0.4889008	−0.2995672	2.001278
<i>Invent</i>	374	−0.0184263	0.7227439	−0.4771927	2.107043
<i>W_kap</i>	374	0.227892	0.1321596	−0.1291738	0.3853835
<i>LT_debt</i>	374	−0.25875	0.0937136	−0.3369235	−0.0512854
<i>Common</i>	374	−0.2400921	0.0449175	−0.2899984	−0.1189817
<i>BPS</i>	374	−0.0079979	0.8949933	−0.8619989	2.910284
<i>Big4</i>	547	0.8665448	0.3403773	0	1
<i>Gender</i>	423	0.9408983	0.2360941	0	1
<i>Mkt_Cap</i>	349	−0.2318855	0.1153687	−0.3439907	0.0572762

Table 2
Pearson correlations.

Short_int	1															
Short_int_adj	0.9857*	1														
KAM	0.0575*	0.0424	1													
ROA	−0.0132	−0.0172	0.1476*	1												
AFee	0.0867*	0.0862*	0.2799*	−0.1230*	1											
Size	0.1374*	0.1340*	0.3434*	−0.0918*	0.7942*	1										
Liab	−0.0461	−0.0455	0.1221*	−0.0984*	0.4482*	0.2819*	1									
W_kap	−0.1222*	−0.1167*	−0.3545*	0.022	−0.6843*	−0.8637*	0.0971*	1								
LT_debt	0.0851*	0.0781*	0.3892*	−0.0941*	0.8158*	0.9959*	0.2684*	−0.8696*	1							
Common	0.1019*	0.0933*	0.4153*	−0.0762*	0.8183*	0.9893*	0.2614*	−0.8974*	0.9904*	1						
BPS	−0.0993*	−0.1097*	−0.0164	−0.1022*	0.1466*	0.1231*	0.6769*	0.1660*	0.1042*	0.1124*	1					
Big4	0.0940*	0.0971*	0.1939*	0.0795*	−0.1711*	0.0636*	0.1189*	−0.0048	0.0829*	0.0818*	0.1175*	1				
Gender	0.1198*	0.1272*	0.1211*	−0.0301	0.1372*	0.0596*	0.0984*	−0.0299	0.0897*	0.0685*	0.1459*	−0.0610*	1			
Mkt_Cap	0.1033*	0.0985*	0.4030*	−0.0729*	0.7668*	0.9945*	0.2423*	−0.8724*	0.9857*	0.9790*	0.0942*	0.0945*	0.1009*	1		

This table illustrates the Pearson Correlations for the regression sample, which includes 2386 firm-year observations between 2010 and 2017. The descriptions of all the variables contained in this table are presented in Appendix. * indicates that the correlation is statistically different from zero with a p-value less than 1%.

Table 3
Pre-post adoption analysis.

Variables	(1)	(2)
	Short_int	Short_int_adj
<i>Post</i>	3.188 (7.732)	3.834 (7.595)
<i>ROA</i>	1.416 (5.991)	1.330 (5.857)
<i>AFee</i>	0.030*** (0.008)	0.028*** (0.008)
<i>Size</i>	−42.891 (25.688)	−41.293 (25.912)
<i>Tangibility</i>	7.306 (10.884)	9.568 (10.978)
<i>Loss</i>	−4.228 (13.751)	−1.848 (13.600)
<i>Invent</i>	−64.238 (69.382)	−51.525 (68.718)
<i>WKap</i>	29.678 (109.624)	6.253 (109.378)
<i>LT_debt</i>	−150.245 (119.738)	−118.402 (119.330)
<i>BV_Eq</i>	288.131 (215.750)	230.385 (215.919)
<i>BPS</i>	39.850 (24.083)	36.651 (23.700)
<i>Big4</i>	280.648*** (91.617)	296.100*** (91.275)
<i>Gender</i>	31.246 (31.625)	28.926 (32.378)
<i>Mkt_Cap</i>	−138.003* (65.536)	−168.705** (68.380)
<i>Constant</i>	−426.790*** (80.177)	−434.292*** (80.472)
<i>Year, Industry & Audit firm FE</i>	YES	YES
Observations	2213	2213
R-squared	0.862	0.853

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

granularity of auditors not being informative for short sellers. Overall, our results are consistent with those of [Gutierrez et al. \(2018\)](#), [Lennox, Schmidt, & Thompson, \(2019\)](#), and [Bedard et al. \(2019\)](#), which suggest that the expanded auditor's report disclosures do not generate capital market reactions.

5.2.3. Unexpected and severe KAM

We build on the approach of [Lennox et al. \(2019\)](#) and construct our measure of unexpected KAM by estimating a model that predicts the number of KAM in each audit report. In the model for predicting expected KAM, we use *ROA*, *Size*, *AFee*, *LT_Debt*, *Big4*, *Gender*, *BPS*, *Loss*, *Invent*, *Tangibility*, and *Mkt_Cap* together with *industry*, *audit firm*, and *year dummies*. We use the error term from this regression to proxy for unexpected KAM. We then re-estimate Equation (3), replacing *KAM* with *Unexpected_KAM*. Results in [Table 8](#) suggest that the coefficients of *Unexpected_KAM* are positive and significant in all estimations. This suggests that the short sellers are likely to react more to the KAM, that are non-standard and can potentially indicate unforeseen risk.

Finally, we test if short sellers' assessments of firm risk are associated with the less generic KAM and signal risks not disclosed by management or the audit committee. We, therefore, select from the KAM sample the issues that we define to be severe and are likely to be associated with higher levels of perceived riskiness. We compute *Severe_KAM* as an indicator variable equal to 1 if the auditors disclose KAM related to fraud, going concern, management override of internal controls, or 0 otherwise. In Models 1 and 2 of [Table 9](#), we find that the coefficients of *Severe_KAM* are positive and significant, suggesting that short interest is positively associated with KAM disclosures perceived as particularly severe and relevant to areas that are more complex to audit.

5.3. Robustness tests

5.3.1. Additional analyses: financial analysts

To determine if the disclosures in the expanded auditor's report impact the decision-making of other sophisticated users of financial statements, we focus on financial analysts' earnings forecasts. Specifically, we use (1) dispersion of analysts' forecasts and (2) analysts' forecast error ([Krishnaswami & Subramaniam, 1999](#)). More specifically, an analyst forecast error is defined as the median analyst EPS

Table 4
Impact of KAM on short interest.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
KAM	0.227* (0.104)	0.193* (0.090)
ROA	−0.175 (0.118)	−0.195 (0.093)
AFee	0.753* (0.346)	0.657 (0.326)
Size	−121.435*** (13.558)	−117.120*** (11.863)
Tangibility	0.325 (0.279)	0.339 (0.261)
Loss	−0.048 (0.376)	−0.053 (0.352)
Invent	1.061 (0.837)	1.093 (0.853)
WKap	18.929*** (2.851)	18.531*** (2.526)
LT_debt	−8.633** (2.607)	−7.791** (2.445)
BV_Eq	92.374*** (10.900)	89.847*** (9.790)
BPS	−1.536*** (0.273)	−1.479*** (0.279)
Big4	4.116*** (0.818)	4.172*** (0.828)
Gender	−1.336* (0.563)	−1.224* (0.455)
Mkt_Cap	−7.998** (2.136)	−7.860** (2.114)
Year, Industry & Audit firm FE	YES	YES
Constant	−4.522*** (0.818)	−4.021*** (0.657)
Observations	1349	1349
R-squared	0.767	0.757

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

forecast for year $t+1$ for all forecasts made within 30 days following the 2013 adoption of the expanded auditor's report less the median consensus forecast of year $t+1$ directly before the 2013 adoption of the expanded auditor's report divided by beginning-of-quarter price.

We test Hypotheses 1–3 using the analyst forecast errors instead of short interest as dependent variables in equation (1). For this analysis, we include firms for which we have complete information on forecast error and dispersion. We also require that the currency used to report actual earnings, forecasted earnings, and share price are identical. To test if the expanded auditor's report has affected the earnings forecasts, we use the following equation:

$$\text{Forecast_err/Dispersion} = \delta_0 + \delta_1 \text{Post} + \delta_2 \text{Adopter} + \delta_3 \text{Post*Adopter} + \delta_4 \text{Firm-specific controls(lagged)} + (\text{Industry, Year, Audit firm dummies}) + \varepsilon_t \quad (3)$$

Results presented in Table 10, Column (1), suggest that dispersion in earnings forecasts is negatively associated with adopting the expanded auditor's report. In turn, we do not find a significant effect in Column (2), where we use forecast error as an outcome variable. In Columns (3) and (4), we find that adopting the expanded auditor's report is related to an incremental decrease in both dispersion in analyst forecast and forecast error. These results suggest that financial analysts largely benefit from the additional information auditors provide in the new audit report.

5.3.2. Market reaction

Prior research finds no short-term market reaction to the 2013 adoption of the expanded auditor's report disclosures (e.g., Burke et al., 2023; Gutierrez et al., 2018; Klevak et al., 2020, 2023; Lennox et al., 2022). We calculate the post-adoption stock return, $POST_CAR$, which is the cumulative abnormal returns (CAR) over the period beginning on day +2 after quarter t 's earnings announcement and ending on the day +1 after the earnings announcement date for quarter $t+1$ pursuant to the 2013 adoption of the expanded auditor's report disclosures. Abnormal returns are measured as the raw daily return from the CRSP minus the return on the portfolio of firms in the same size and book-to-market deciles.

Table 5
Impact of Words_KAM on short interest.

Variables	(1)	(2)
	Short_int	Short_int_adj
Words_KAM	0.041 (0.267)	0.072 (0.268)
ROA	−0.163* (0.064)	−0.173* (0.066)
AFee	1.012*** (0.192)	0.892** (0.207)
Size	0.544* (0.215)	0.546* (0.219)
Tangibility	0.106 (0.293)	0.087 (0.290)
Loss	2.693** (0.727)	2.543** (0.862)
Invent	17.596** (6.298)	17.425** (5.595)
Wkap	−8.738** (2.798)	−8.007** (2.731)
LT_debt	98.970*** (13.644)	96.251*** (11.754)
BV_Eq	3.520 (2.132)	2.739 (1.813)
BPS	−1.939*** (0.384)	−1.839*** (0.390)
Big4	1.924*** (0.395)	1.893*** (0.379)
Gender	−1.346*** (0.183)	−1.252*** (0.173)
Mkt_Cap	−8.241* (3.156)	−8.069** (2.667)
Year, Industry & Audit firm FE	YES	YES
Constant	−0.452 (0.612)	−0.593 (0.565)
Observations	1251	1251
R-squared	0.770	0.755

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

$$\text{CAR/POST_CAR} = \delta_0 + \delta_1 \text{Post} + \delta_i \text{Firm-specific controls(lagged)} + (\text{Industry, Year, Audit firm dummies}) + \text{et} \quad (4)$$

To assess the robustness of previous findings for CAR computed in a long-term window around the adoption of the expanded auditor's report, we replicate the main tests of [Gutierrez et al. \(2018\)](#), [Lennox et al. \(2022\)](#), and [Burke et al., 2023](#) and find consistent results of no market reactions to KAM disclosures. Specifically, we cannot detect a sizeable effect of the expanded auditor's report adoption for short and long-window CAR. Our investigation uncovers insignificant coefficients concerning KAM disclosures in [Table 11](#) (Columns 1 and 2). Analyzing the coefficient associated with the KAM variable in relation to CAR yields insignificant outcomes.

Consequently, we proceed with further investigations using CAR to explore potential information content during the (POST_CAR) quarter's earnings announcement of KAM. In this context, the term "KAM already disclosed" denotes that we have confirmed that the company unveiled the CAR before the auditor's disclosure for (POST_CAR) disclosures quarter t's earnings announcement. We examine market responses for supplementary analysis by observing the three-day CAR centered around an earnings announcement. An additional test was performed to further validate the pre-existing knowledge of KAM information within the market. Consequently, in our analysis (POST_CAR) disclosures quarter t's earnings announcement, we investigate whether the CAR is associated with the first KAM disclosures the firm had already divulged. We anticipate an earlier revelation of these CAR could result in an insignificant overlap with the disclosures earnings announcement. To elucidate, if the firm had already publicly disclosed CAR related to revenue recognition (such as applying a complex percentage of completion method), a similar disclosure by the auditor would lead to a substantial redundancy, even though the specific CAR was not disclosed by the auditor previously. Therefore, our investigation is focused on the KAMs that firms had already disclosed in the initial year disclosures quarter's earnings announcement before auditors included them in the extended audit reports. Our pursuit of the "already disclosed" KAMs commences by thoroughly analyzing each firm's year t-1 disclosures quarter t's earnings announcement alongside the preceding CAR records. These findings are consistent with the results reported by [Gutierrez et al. \(2018\)](#), [Reid et al. \(2019\)](#), [Lennox et al. \(2022\)](#) and [Burke et al., 2023](#).

5.3.3. Alternative sample periods

In our main analysis, we follow [DeFond, Hung, Li, and Li \(2015\)](#) and use an event window of two years around the mandatory adoption of the expanded auditor's report to limit the effect of other potential confounding events. We repeat our pre-post analysis by

Table 6
Impact of Average_Words_KAM on short interest.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
Average_Word_KAM	−0.151 (0.243)	−0.105 (0.074)
ROA	−0.290** (0.086)	−0.292*** (0.058)
AFee	0.626** (0.234)	0.549 (0.311)
Size	0.261 (0.154)	0.279 (0.240)
Tangibility	0.070 (0.436)	0.036 (0.268)
Loss	1.099** (0.374)	1.093 (0.915)
Invent	16.481** (4.721)	16.731** (3.890)
Wkap	−8.278*** (1.520)	−7.508** (2.292)
LT_debt	83.549*** (12.105)	82.900*** (14.369)
BV_Eq	8.581*** (1.671)	7.451** (2.456)
BPS	−1.412*** (0.334)	−1.377** (0.398)
Big4	10.773*** (1.281)	10.375*** (1.815)
Gender	−1.034** (0.358)	−0.933* (0.355)
Mkt_Cap	−7.252* (3.060)	−7.396** (2.545)
Year, Industry & Audit firm FE	YES	YES
Constant	−10.830*** (1.614)	−9.954** (2.427)
Observations	1349	1349
R-squared	0.762	0.752

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

(1) using a one-year window around the adoption and (2) dropping the transition year 2013. The coefficient of *Post* remains insignificant for all estimations, suggesting that the effect of the adoption is not due to confounding effects concurrent with the 2013 regulation implementation.

5.3.4. The impact of corporate governance

One concern in our tests can potentially be connected to the causal relations between expanded auditor's report disclosures and short-selling interests. Specifically, our results can be impacted by the effect of corporate governance variations on short sellers' valuation decisions. To address this concern, we use the ASSET4 database and extract yearly overall corporate governance scores for a subsample in our data. The ASSET4 database of Thomson Reuters' Research Data provides reliable firm-level scores on economics, environmental, social, and governance pillars (Ioannou & Serafeim, 2012). We further include the scores in the models and find that our results are largely unaffected.

5.3.5. The impact of audit committee disclosures

Aside from adopting the expanded auditor's report, the FRC mandated for fiscal years after 2013 that the audit committee also make disclosures regarding the main KAM. Since these disclosures are made concurrently with the ones of auditors in the expanded auditor's report, they can potentially impact the risk assessment and firm valuation made by short sellers. To address this issue, we use the audit committee disclosure data made public by Gutierrez et al. (2018) and include it in our models. Untabulated results show that when we add the disclosures of the audit committee in all models, our results remain unchanged.

5.3.6. The impact of ownership

We consider the effect of institutional ownership (*Inst_Own*) on our measures of short-selling interest since prior research has shown that institutional ownership is positively associated with short-selling (Bushee, 1998; D'Avolio, 2002; Asquith et al., 2005). When including *Inst_Own* in all our estimations, the results of our main tests remain unchanged.

Table 7
Impact of materiality thresholds on short interest.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
Materiality	0.284 (0.211)	0.251 (0.187)
ROA	−0.345* (0.181)	−0.341* (0.166)
AFee	0.404 (0.289)	0.354 (0.296)
Size	0.164 (0.187)	0.200 (0.191)
Tangibility	0.106 (0.423)	0.082 (0.394)
Loss	0.776 (0.673)	0.847 (0.733)
Invent	18.102*** (5.136)	17.820*** (4.874)
WKap	−8.164*** (1.501)	−7.383*** (1.012)
LT_debt	89.222*** (20.577)	87.231*** (19.486)
BV_Eq	10.080** (4.183)	8.604* (4.219)
BPS	−1.493*** (0.350)	−1.445*** (0.320)
Big4	3.824*** (0.440)	3.936*** (0.467)
Gender	−0.463 (0.377)	−0.469 (0.366)
Mkt_Cap	−8.119** (3.412)	−7.955** (3.078)
Year, Industry & Audit firm FE	YES	YES
Constant	−2.285 (1.907)	−1.988 (1.810)
Observations	1349	1349
R-squared	0.763	0.754

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

5.3.7. Heckman two-stage model

We follow [Blundell and Powell \(2007\)](#) to use a two-stage Heckman procedure that addresses the effect of endogeneity in our data ([Bouslah, Kryzanowski, & M'Zali, 2018](#); [Lennox, Francis, & Wang, 2012](#)). Specifically, we employ the [Heckman \(1979\)](#) approach and use a two-stage estimation procedure by computing the Inverse Mills ratio. After computing the Inverse Mills ratio in the first stage, we include it in the second stage as a control variable in our main estimations. Untabulated regression results show that the direction and significance of our variables of interest remain the same. Consequently, endogeneity concerns do not significantly affect the results of our main tests in [Tables 3–9](#).

5.3.8. Quasi neutral experiments

In this section, we use an alternative method of a quasi-natural experiment to further address potential endogeneity issues in our data. Following, [Zhai et al. \(2021\)](#), we employ a quasi-natural experiment in KAM disclosures spanning from 2010 to 2017. We perform quasi-natural experiments to ensure the reliability of our results presented in [Tables 1–9](#). The treatment group (dual-listed firms) is identified by the variable Treat, representing firms complying with the new audit standards in 2013 (coded as one) and zero otherwise. The POST variable, coded as one for fiscal years ending in or after 2013 and zero otherwise, indicating the period post-adoption. Furthermore, indicators for a high number of reported KAMs ($KAM \geq 3$) and a low number ($KAM \leq 2$) are incorporated for additional analyses. The coefficients for Treat * POST, POST * $KAM \geq 3$, and POST * $KAM \leq 2$ provide evidence of the interaction effects between treatment status, post-adoption period, and the number of reported KAMs on short interest and short interest adjusted. The findings suggest that the treatment group (firms complying with new audit standards) experienced a decrease in short interest post-adoption, especially for firms with a low number of reported KAMs ($KAM \leq 2$) compared to firms with a higher number of reported KAMs ($KAM \geq 3$). Results presented in [Table 12](#) confirm the robustness of our research methodology and the reliability of our findings consistent with [Zhai et al. \(2021\)](#).

Table 8
Impact of Unexpected KAM on short interest.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
Unexpected_KAM	0.115*** (0.023)	0.109*** (0.023)
ROA	0.092*** (0.029)	0.089*** (0.030)
AFee	1.490*** (0.168)	1.399*** (0.171)
Size	9.003*** (2.864)	11.028*** (3.140)
Tangibility	0.341*** (0.082)	0.382*** (0.083)
Loss	0.118 (0.083)	0.171** (0.086)
Invent	0.279 (0.334)	0.440 (0.335)
Wkap	−1.904*** (0.609)	−2.366*** (0.631)
LT_debt	−5.143*** (0.849)	−5.100*** (0.841)
BV_Eq	−2.144 (1.530)	−3.795** (1.717)
BPS	0.134* (0.060)	0.108* (0.060)
Big4	−0.619*** (0.176)	−0.630*** (0.180)
Mkt_Cap	−4.179*** (0.529)	−4.491*** (0.599)
Year, Industry & Audit firm FE	YES	YES
Constant	2.095*** (0.348)	2.308*** (0.360)
Observations	1210	1210
R-squared	0.631	0.632

Robust standard errors in parentheses. All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

6. Conclusion

To improve the effectiveness of audit reports, global regulators and standard setters require auditors to communicate the audit scope, materiality thresholds, and KAM to investors. We examine the association between the expanded auditor's report disclosures and short interest. Short sellers, as informed, investors are interested in the information value of disclosed KAM. We posit that short sellers, based on their prediction ability and skilled information processing ability, can anticipate information about accounts with high levels of judgement and financial reporting risks prior to KAM reporting. Using a large sample of the premium-listed UK firms in 2010–2017, we find that short-selling interest is not significantly associated with the 2013 adoption of the expanded auditor's report. Our cross-sectional tests in the post-2013 period further indicate that the attributes of the expanded audit report have a largely insignificant effect on short-term interest. However, we find marginal evidence that short sellers incorporate the information the number of KAM provides into their investment decisions. This finding aligns with investors perceiving auditors' risk assessments as increasing the firm's overall riskiness.

Furthermore, we examine the association between unexpected and severe KAM and short interest and find that disclosed unexpected and severe KAM is significantly and positively associated with short interest. In our capital market reaction analysis, we cannot detect a sizeable effect of the expanded auditor's report adoption for short and long-window KAM. However, financial analysts largely benefit from the additional information auditors provide in the new audit report. The valuation implication of KAM disclosures for short sellers and financial analysts supports global regulators' and standard setters' initiatives to improve the information value of audit reports. Overall, our results support both prediction ability and skilled information processing ability theories that short sellers have a superior ability to identify firms with financial reporting challenges, including KAM disclosures of high levels of judgement and financial reporting risks prior to KAM reporting and to take short positions before these challenges are publicly disclosed.

This study contributes to the auditing literature in several ways. First, we extend the research on expanded auditor's report disclosures by examining short sellers' reactions to KAM disclosures. Our findings suggest that short sellers already consider the information provided by auditors before the public disclosure of the audit report, supporting the prediction and skilled information processing theories of short sellers. Thus, future research may use the differential analysis based on responses from sophisticated investors to expand understanding of auditor behavior and audit outcomes as short sellers perceive. Second, we provide direct evidence that KAM reporting assesses the usefulness of the expanded auditor's report disclosures from the short seller's perspective. Thus, results present practical implications for investors and audit clients given the value relevance of unexpected and severe KAM

Table 9
Impact of Severe KAM on short interest.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
<i>Severe_KAM</i>	3.765*** (0.362)	3.821*** (0.368)
<i>ROA</i>	−0.081*** (0.026)	−0.086*** (0.028)
<i>AFee</i>	3.367*** (0.269)	3.321*** (0.279)
<i>Size</i>	−1.218*** (0.139)	−1.199*** (0.143)
<i>Tangibility</i>	0.323*** (0.079)	0.363*** (0.080)
<i>Loss</i>	0.077 (0.091)	0.124 (0.093)
<i>Invent</i>	−0.820*** (0.251)	−0.646** (0.257)
<i>Wkap</i>	−1.411** (0.654)	−1.714** (0.674)
<i>LT_debt</i>	−2.537*** (0.391)	−2.081*** (0.425)
<i>BV_Eq</i>	4.544*** (0.691)	3.737*** (0.739)
<i>BPS</i>	1.381*** (0.112)	1.354*** (0.116)
<i>Big4</i>	10.830*** (1.046)	11.286*** (1.060)
<i>Gender</i>	3.402*** (0.259)	3.400*** (0.265)
<i>Mkt_Cap</i>	−4.196*** (0.528)	−4.587*** (0.686)
<i>Year, Industry & Audit firm FE</i>	YES	YES
Constant	−14.468*** (1.429)	−14.915*** (1.447)
Observations	1213	1213
R-squared	0.723	0.716

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

Table 10
Additional test - Impact of expanded auditor's report on financial analysts' earnings forecasts.

VARIABLES	(1)	(2)	(3)	(4)
	<i>Dispersion</i>	<i>Forecast_Err</i>	<i>Dispersion</i>	<i>Forecast_Err</i>
<i>Post</i>	−0.053* (0.028)	−0.196 (0.133)	0.054 (0.042)	−0.069 (0.120)
<i>KAM</i>			0.049 (0.035)	0.054 (0.042)
<i>Post * KAM</i>			−0.107*** (0.041)	−0.127** (0.063)
<i>BV_Eq</i>	−0.013 (0.008)	−0.021 (0.018)	−0.013 (0.008)	−0.021 (0.018)
<i>Size</i>	−0.012* (0.008)	−0.034* (0.018)	−0.012 (0.008)	−0.034* (0.018)
<i>Ebex</i>	0.001 (0.005)	0.018 (0.016)	0.001 (0.005)	0.018 (0.016)
<i>ROA</i>	−0.648** (0.291)	−1.441* (0.811)	−0.648** (0.292)	−1.441* (0.812)
<i>Constant</i>	0.010 (0.061)	0.318 (0.210)	−0.039 (0.073)	0.264 (0.203)
<i>Year, Industry & Audit firm FE</i>	YES	YES	YES	YES
Observations	870	870	870	870
R-squared	0.041	0.065	0.041	0.065

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

Table 11
Market reaction.

<i>Variables</i>	(1)	(2)
	<i>CAR</i>	<i>POST CAR</i>
<i>Post</i>	−1.113 (−1.159)	−1.223 (−0.875)
<i>KAM</i>	−0.131 (−0.650)	−0.124 (−0.593)
<i>Post * KAM</i>	−1.105 (−0.530)	−1.111 (−1.240)
<i>AFee</i>	2.156*** (0.258)	2.631*** (0.388)
<i>Size</i>	0.031*** (2.121)	0.144*** (2.088)
<i>Tangibility</i>	0.141*** (3.103)	0.513*** (3.176)
<i>Loss</i>	0.087 (0.082)	0.135 (0.081)
<i>Invent</i>	−0.129 (−0.725)	−0.430 (−0.750)
<i>Wkap</i>	−2.802*** (0.405)	−2.715*** (0.322)
<i>LT_debt</i>	−2.221*** (0.347)	−4.220*** (0.413)
<i>BV_Eq</i>	−3.273** (2.420)	−2.514** (2.625)
<i>BPS</i>	0.213** (0.052)	0.216* (0.031)
<i>Big4</i>	9.630*** (1.213)	8.484*** (1.141)
<i>Gender</i>	2.203*** (0.315)	2.320*** (0.374)
<i>Mkt_Cap</i>	−3.162*** (0.313)	−3.353*** (0.324)
<i>Year, Industry & Audit firm FE</i>	YES	YES
Constant	0.031 (0.058)	0.546 (1.046)
Observations	870	870
R-squared	0.026	0.016

Robust standard errors in parentheses.

All variables are defined in Appendix.

***p < 0.01, **p < 0.05, *p < 0.1.

disclosures. Third, our results provide policy implications and contribute to the global debate on the value relevance of KAM disclosures by showing that only unexpected and severe KAM disclosures exhibit value relevance. Finally, our results contribute to the extant research on KAM/CAM reporting by supporting the unexpected and severe KAM disclosures that investors and auditors view as relevant and material.

Our results should be interpreted with caution because of the potential limitations. First, our short-selling data is limited by the availability in Compustat and had-collected approach. The meticulous hand-collection of data for the expanded auditor's report would have presented significant challenges beyond the chosen 2010–2017 period. Several limitations of hand-collected data are: (1) limited scope of collecting data manually can be time-consuming and resource-intensive and thus the scope of the data may be narrower compared to automated collection methods; (2) subjectivity as hand-collected data can be subjective and thus introduce variability and reduce the reliability of the data; and (3) bias and inaccuracy that can stem from the context in which the data is collected, personal beliefs or perspectives. Despite these limitations in our hand-collected data, we took steps to mitigate them. Future studies should try enlarging our dataset by tapping into more comprehensive sources of short-selling data and hand-collected KAM data for more years before and after the adoption of KAM regulations. Second, despite our efforts to increase the robustness of our estimations, we cannot fully capture variations in the private information sets available to short sellers in our tests. However, firmly believe that the Heckman model and quasi-neutral experiments, used in our paper, are well-suited to tackle sample selection bias in cross-sectional data. As CAM disclosures become available in the United States, future research should investigate the value-relevance of these disclosures.

Data availability

Data will be made available on request.

Table 12
Additional test: Quasi neutral experiments.

Variables	(1)	(2)
	<i>Short_int</i>	<i>Short_int_adj</i>
<i>Treat</i> * <i>POST</i>	−1.105 (−0.530)	−1.085 (−0.490)
<i>POST</i> * <i>KAM</i> ≥ 3	−0.127 (−1.063)	−0.116 (−1.032)
<i>POST</i> * <i>KAM</i> ≤ 2	−0.107* (−0.341)	−0.097* (−0.291)
<i>Treat</i>	0.227 (0.570)	0.196 (0.480)
<i>POST</i>	3.188 (7.732)	3.157 (7.621)
ROA	−0.083*** (0.039)	−0.073*** (0.023)
AFee	3.234*** (0.261)	3.121*** (0.243)
Size	−1.124*** (0.147)	−1.176*** (0.132)
Tangibility	0.318*** (0.074)	0.291*** (0.061)
Loss	0.043 (0.083)	0.113 (0.072)
Invent	−0.737*** (0.431)	−0.698** (0.322)
Wkap	−1.324** (0.536)	−1.215** (0.493)
LT_debt	−2.237*** (0.312)	−2.172*** (0.297)
BV_Eq	4.421*** (0.597)	3.987*** (0.538)
BPS	1.337*** (0.134)	1.323*** (0.113)
Big4	10.189*** (1.035)	10.165*** (1.020)
Gender	3.208*** (0.221)	3.197*** (0.216)
Mkt_Cap	−4.365*** (0.596)	−4.297*** (0.571)
Year, Industry & Audit firm FE	YES	YES
Constant	−15.271*** (1.367)	−15.199*** (1.246)
Observations	1213	1213
R-squared	0.672	0.644

***p < 0.01, **p < 0.05, *p < 0.1.

Appendix

Variables and their Definition and Sources

Firm-specific variables	Source
<i>Short_int</i>	Total shares sold short (positions held on the 15th business day of each month) scaled by the total shares outstanding at time <i>t</i> ; <i>Compustat's Supplemental Short Interest File</i>
<i>Short_int_adj</i>	Total adjusted shares sold short (positions held on the 15th business day of each month) scaled by the total shares outstanding at time <i>t</i> ; <i>Compustat's Supplemental Short Interest File</i>
ROA	The ratio of earnings before interest and taxes to total assets at the end of the most recent fiscal year; <i>DataStream</i>
Size	The natural logarithm of total assets; <i>DataStream</i>
AFee	The amount of audit fees paid to external auditors; <i>Hand-collected - annual reports</i>
LT_debt	The ratio of total long-term debts to total assets; <i>DataStream</i>
BPS	The ratio of the firm's book value divided by the total number of shares outstanding; <i>DataStream</i>
Big4	Indicator variable equal to 1 for Big4 audit companies or 0 otherwise; <i>DataStream</i>
Loss	Indicator variable equal to 1 if net income is negative or 0 otherwise; <i>DataStream</i>
Wkap	The firm's working capital; <i>DataStream</i>
Mkt_Cap	The number of shares outstanding multiplied by the price share; <i>DataStream</i>

(continued on next page)

(continued)

Firm-specific variables		Source
<i>Inst_Own</i>	The percentage of shares owned by institutional investors;	<i>Boardex</i>
<i>Invent</i>	The ratio of total inventories to total assets;	<i>DataStream</i>
<i>Tangibility</i>	The ratio of current assets to current liabilities.	<i>DataStream</i>
<i>BV_Eq</i>	The book value of common equity;	<i>DataStream</i>
Audit Report-Specific Variables		
<i>KAM</i>	The total number of KAMs disclosed in the expanded auditor's report;	<i>Hand-collected - annual reports</i>
<i>Severe_KAM</i>	Indicator variable equal to 1 for KAM that relate to fraud, going concern, and management override of internal controls or 0 otherwise;	<i>Hand-collected - annual reports</i>
<i>Materiality</i>	The natural logarithm of the materiality amount disclosed in the expanded auditor's report;	<i>Hand-collected - annual reports</i>
<i>Words_KAM</i>	The natural logarithm of the number of words in KAM paragraph;	<i>Hand-collected - annual reports</i>
<i>Average Words_KAM</i>	The natural logarithm of the number of words in the KAM paragraph. To calculate average number of words, we divide the total number of words of the KAM paragraph in the expanded auditor's report by the total number of KAMs	<i>Hand-collected - annual reports</i>
<i>Pre-Post Dummy</i>		
<i>Post</i>	An indicator variable equal to 1 for all firms with a fiscal year-end on and after October 1, 2013, or 0 otherwise.	

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